How much does product classification cost in basis points?

Multilingual classification of Thai product text for consumer price measurement

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Abstract

[This part is written once the gold labels are evaluated.]

JEL codes: C55, C81, E31. **Keywords:** consumer price index, product classification, multilingual text embeddings, large language models, web prices, measurement error, Thailand.

1 Introduction

[This part is written once the gold labels are evaluated.]

2 Related work

Online prices. Prices collected from retailers' websites have become a measurement source in their own right (Cavallo and Rigobon, 2016). For large multi-channel retailers, online and offline prices are largely the same (Cavallo, 2017), and online prices help forecast and nowcast inflation (Aparicio and Bertolotto, 2020; Macias et al., 2023). Every such index needs each product assigned to a category of the official basket first; this paper is about that step.

Classification in official statistics. Eurostat's guides on scanner data and web scraping for the HICP treat classification as one of the main processing steps (Eurostat, 2017, 2020), and euro-area statistics have moved towards these sources (Eiglsperger, 2019). National statistical offices have classified products with machine learning: the ONS with clustering and decision trees (Metcalfe et al., 2016) and with semi-supervised label propagation into CPIH basket items (Martindale et al., 2020), Statistics Netherlands with supervised classifiers, of which logistic regression performed best (Harms and Spinder, 2019), and the U.S. Census Bureau with text classification of product descriptions for the Economic Census (Roberson, 2021). The Australian Bureau of Statistics reported that it was investigating machine learning to automate the classification of web-scraped products (Australian Bureau of Statistics, 2020), and a UNECE project surveyed such uses across offices (United Nations Economic Commission for Europe, 2021). These studies report accuracy; they do not translate it into the error of the resulting index.

Project Spectrum. The closest work is the BIS Innovation Hub's Project Spectrum (BIS Innovation Hub, 2026), which classifies the ECB's Daily Price Dataset into ECOICOP 2018 subclasses. It finds that text embeddings with classic classifiers reach accuracy comparable to LLM prompting at a small fraction of the cost and time (BIS Innovation Hub, 2026, p. 3), and reports accuracy weighted by CPI expenditure at the subclass level (BIS Innovation Hub, 2026, pp. 22–24). It argues that, for nowcasting,

*The views expressed are the author's own and do not necessarily reflect those of the Bank of Thailand. This work uses only publicly available data. Code and derived data: `https://github.com/thanatchon36/cpilens`.

errors at the subclass level usually matter less than errors at higher levels of the hierarchy, depending on how differently the confused categories' prices move (BIS Innovation Hub, 2026, p. 25), and it names two next steps: testing the approach on other data sets and languages, and building CPI indices from the classified data to back-test them against official inflation (BIS Innovation Hub, 2026, p. 30). This paper takes up both for one non-European language and turns the second into a measurement: the error a given confusion matrix adds to measured inflation, in basis points. Our setting differs from Spectrum's in taxonomy, language, product universe and labels, so the numbers are not directly comparable.

Models. We use multilingual sentence encoders (Reimers and Gurevych, 2019; Wang et al., 2024; Chen et al., 2024) and a fine-tuned multilingual encoder (Conneau et al., 2020). LLMs have been shown to match or beat crowd workers on text annotation (Gilardi et al., 2023), and cascades that send only hard items to an expensive model can cut cost at little loss of accuracy (Chen et al., 2023). Calibrated confidence (Guo et al., 2017) makes selective classification possible, trading coverage for accuracy (Geifman and El-Yaniv, 2017); we weight both axes of that trade-off by CPI expenditure.

3 Data

All data are public. Table 1 in Appendix A lists the sources and their licences; the repository records the retrieval date and checksum of every file.

3.1 The Thai CPI

The Thai consumer price index is compiled monthly by the Trade Policy and Strategy Office (TPSO) of the Ministry of Commerce. The current base year is 2566 in the Buddhist era (2023), and the series on this base has been in use since January 2568 (2025) (Trade Policy and Strategy Office, 2025). Like most national CPIs it is a Lowe index (International Monetary Fund et al., 2020): expenditure weights from a base period, price-updated to a price reference month, here December 2567. TPSO's open API publishes monthly indices for the headline, divisions, groups and items (Trade Policy and Strategy Office, 2026). It publishes no weights. The monthly release publishes price-updated weight shares for the official aggregates only (headline, the divisions, the food groups and the analytical series such as core CPI). We recover those shares from the releases and check that they add up the hierarchy and reproduce the official headline from the published sub-indices.

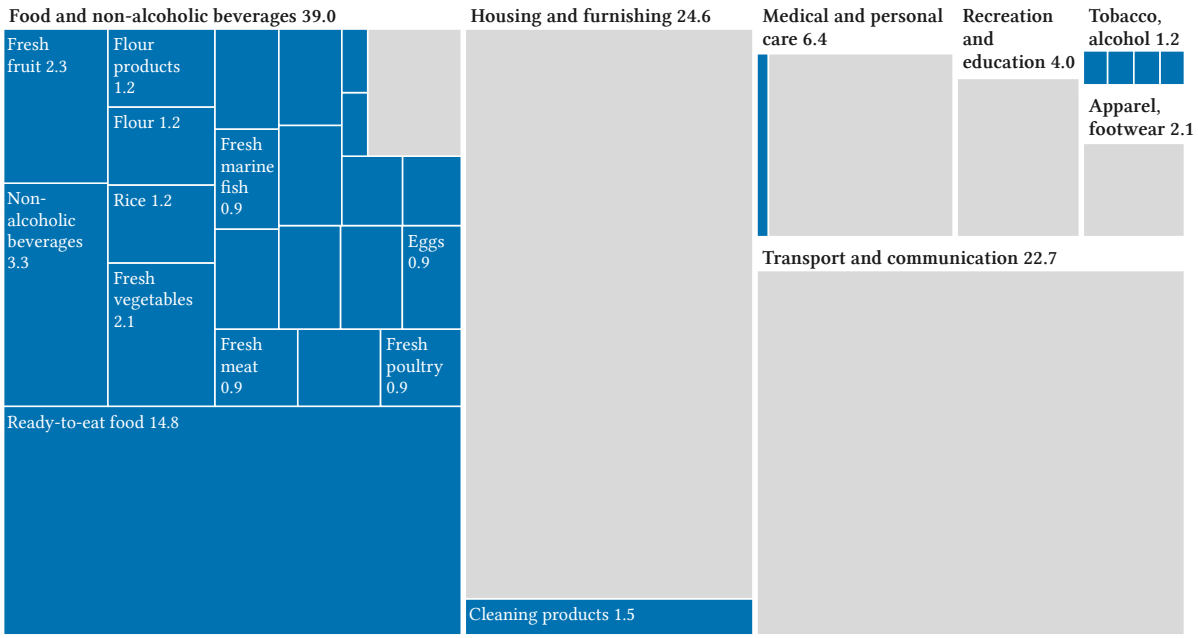
Below the published aggregates, item weights are not public. We partition the item codes into *cells*: each official aggregate owns the items whose deepest published ancestor it is, and the cell weight is the aggregate's weight minus that of the published aggregates nested below it. Within a cell we estimate item weights in two ways and report both: an equal split (primary) and non-negative least squares on the Lowe identity, which chooses the split that best reproduces the cell's index from its items' indices (sensitivity). Neither is an official weight, and the paper never calls it one.

3.2 Scope: the items a product classifier can reach

A classifier of packaged-product text can only measure items sold as packaged products. We define 28 in-scope CPI items: food at home, ready-to-eat food, non-alcoholic beverages, alcoholic beverages, tobacco, personal care and cleaning products. Each is mapped to its COICOP 2018 class (United Nations Statistics Division, 2018). Everything else (rents, energy, transport, services, clothing, durable goods) is out of scope (OOS). In-scope items carry 40.5% of the headline weight under the equal split and 43.5% under NNLS, most of it food (Figure 1).

3.3 Product text

Open Food Facts. Open Food Facts (OFF) is a crowd-sourced, openly licensed database of packaged products (Open Food Facts contributors, 2026). We use its Parquet export and that of its sister project



Source: TPSO consumer price indices and weights (base 2566); author's calculations.

Note: Area: price-updated CPI weight (% of headline). Blue: the 28 items in the classifier's scope; grey: the rest of each division. The 16 smallest are unlabelled; the appendix lists every item. Item weights inside an official weight cell are an equal split (estimated).

Figure 1: Packaged-product items are a large share of the Thai CPI basket, and most of it is food. Area: price-updated CPI weight (% of headline), by division; blue, the in-scope items; grey, the rest of each division. Item weights inside an official weight cell are an equal split (estimated). Source: TPSO; author's calculations.

Open Beauty Facts. Our extract holds 76,269 products from 18 markets: Thailand, other Asian markets and a sample of European products. Of the 10,526 products tagged as sold in Thailand, 7,303 have a name in Thai script. Model input is the product's name, brand and quantity, joined as `name | brand | quantity`.

Weak labels. OFF products carry crowd-sourced category tags. A hand-curated crosswalk maps tags to CPI items, following TPSO's own item headings rather than intuition about COICOP (for example, snacks belong to flour products in the Thai CPI). This gives weak labels: 52,255 non-Thai products and 3,945 Thai products. Weak labels are used for training and development, never for evaluation.

Official specifications. The Ministry of Commerce publishes the product specifications TPSO uses for price collection: short Thai descriptions with brand and pack size, each under an official item code (Ministry of Commerce, Thailand, 2026). These give 31,822 official labels. They are not shop listings, but they are the closest public analogue of labelled Thai product text. We split them by item code, so that specifications of one item never straddle training and test: 25,805 for training and validation, 6,017 for test.

3.4 Gold labels

Evaluation uses a gold sample of Thai OFF products labelled by the author. The frame is the 7,303 Thai-market products with a Thai-script name. The sample of 600 products is stratified by the CPI weight cell of the product's weak label (13 strata, including products with no weak label), with a floor per stratum and the rest allocated half by stratum size and half by cell weight, so heavily weighted items

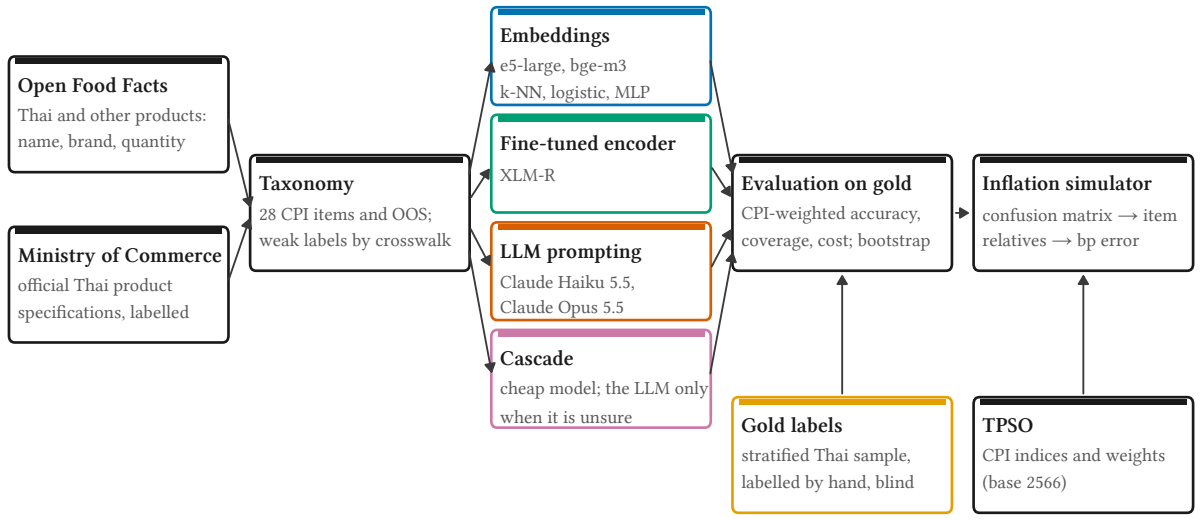


Figure 2: The pipeline runs from public product text to basis points of measured inflation. Data sources (left) feed a taxonomy and weak labels, four families of classifier, an evaluation on hand-labelled gold data and the inflation simulator. Source: author's design.

are well represented. Inclusion probabilities are stored and every gold estimate is design weighted. Gold products, and any training text identical to a gold text, are removed from every training pool.

The labelling tool shows the product's names, brand, quantity and front image, and a search over the Thai and English item names. It never shows a model's prediction, the weak label or the stratum. Besides the items and OOS, the labeller can answer "cannot tell from the information shown" (UNK, excluded from accuracy and counted) and flag any label as unsure. A random 150 of the sample is labelled a second time, blind, in a different order and at least 48 hours after the first label. Because both passes are by the same person, their disagreement measures intra-annotator consistency, a lower bound on the disagreement between two annotators.

4 Methods

Figure 2 summarises the pipeline. Every method maps a product text to one of the 28 in-scope items or OOS, with a calibrated confidence; some may abstain.

4.1 Classifiers

Frozen multilingual embeddings. Following Spectrum's design (BIS Innovation Hub, 2026, pp. 3, 19–20), the first family embeds each text once and trains a light classifier on the vectors. We use two open multilingual embedding models run locally, e5-large (`intfloat/multilingual-e5-large`; Wang et al., 2024) and bge-m3 (`BAAI/bge-m3`; Chen et al., 2024). On top of each we fit three classifiers: cosine k-nearest neighbours with similarity-weighted votes, $k \in \{1, 3, 5, 10, 20, 50\}$ chosen on validation macro recall; multinomial logistic regression; and a one-hidden-layer perceptron (512 units, dropout 0.2). The last two share one trainer: Adam with learning rate 10^{-3} and weight decay 10^{-4}, class weights proportional to the inverse square root of class frequency, and early stopping on validation loss (patience 3).

Fine-tuned encoder. XLM-R base (`FacebookAI/xlm-roberta-base`; Conneau et al., 2020) with a classification head, fine-tuned end to end on the same training pools with the same class weights and early stopping (4 epochs at most, batch 64, AdamW at 3×10^{-5} with linear warm-up and decay, inputs truncated to 64 tokens).

LLM prompting. Two models of different cost, Claude Opus 5.5 and Claude Haiku 5.5, through the Anthropic API, one product per request. A fixed system prompt gives the instructions, the item list grouped by CPI group with short descriptions grounded in TPSO's item headings, and 2 official specifications per item drawn from the training split (Appendix C). The answer is constrained by a JSON schema: the model names the group, then an item inside it, and a verbalised confidence, the probability in per cent that the item is right. Refusals count as abstentions. Every response is cached and every billed call is logged with its tokens and cost; total spend was capped at USD 60 and came to USD 5.62.

Cascade. A cheap classifier answers when its calibrated confidence is at least a threshold t; otherwise the product is routed to an LLM. We pair the cheap classifiers with each LLM. For each pair, t is the threshold with the highest weighted accuracy on the Thai weak-label validation split, the development set closest to the gold population, with ties broken by lower cost. It is fixed before gold is touched, and gold then scores that one threshold.

Training data and the learning curve (RQ1). Each embedding classifier is trained on a sequence of pools: non-Thai weak labels only (zero-shot cross-lingual transfer to Thai); plus Thai weak labels; plus $n \in \{100, 250, 500, 1{,}000\}$ or all official Thai labels; and official labels alone. The budget n counts every official label a run touches, validation included, so a small-budget run cannot tune on thousands of extra official labels.

Calibration. Each run's logits are divided by one scalar temperature fitted on its own validation rows by negative log-likelihood (Guo et al., 2017); for k-NN the logits are log vote shares. The calibrated top-class probability is the confidence used by the coverage frontier and the cascade.

4.2 Evaluation

Expenditure-weighted accuracy. What matters for a price index is not how many products are classified correctly but how much CPI weight is measured correctly. Let w_k be the CPI weight of item k and r_k the recall of item k over the Thai product universe, estimated from the gold sample with Horvitz–Thompson weights $1/\pi_i$ (Horvitz and Thompson, 1952). The headline metric is

$$\mathrm{acc}_w = \frac{\sum_k w_k r_k}{\sum_k w_k}, \qquad r_k = \frac{\sum_{i:\, y_i = k} \mathbb{1}[\hat{y}_i = k]/\pi_i}{\sum_{i:\, y_i = k} 1/\pi_i}, \tag{1}$$

the CPI-weighted mean recall over in-scope items, a weighted version of Spectrum's in-class recall (BIS Innovation Hub, 2026, p. 22). We also report unweighted accuracy, which counts OOS products, and macro-F1. Abstentions are errors at full coverage.

Coverage, cost and uncertainty. Sweeping a confidence threshold traces a weighted accuracy–coverage frontier (Geifman and El-Yaniv, 2017): the CPI-weighted share of products answered against acc_w on those. Cost per million products is the measured mean API cost per call plus local compute at an assumed USD 1.00 per GPU-hour times measured seconds per item. All intervals come from 1,000 stratified bootstrap resamples of the gold sample (Efron and Tibshirani, 1993). Method comparisons are paired: every method is scored on the same resamples, and the comparisons were fixed in code before any gold label existed.

4.3 From confusion matrices to basis points (RQ3)

Setting. For an aggregate A with member items i, the Lowe index satisfies

$$\frac{I_{A,t}}{I_{A,L}} = \sum_{i \in A} \frac{w_i}{W_A} R_{i,t}, \qquad R_{i,t} = \frac{I_{i,t}}{I_{i,L}}, \tag{2}$$

where L is the price reference month, w_i are price-updated weights and $W_A = \sum_{i \in A} w_i$. We take the official item relatives $R_{i,t}$ as the true price movement of every product in item i. This isolates the error due to classification: a real web-price index would add sampling, matching and quality-adjustment error on top.

Misclassification. Let n_k be the universe share of products whose true item is k and P a method's confusion matrix, P_{kj} the probability that a product of item k is classified as j. Item j is then measured from a mixture of the products sent to it:

$$\tilde{R}_{j,t} = \sum_k \phi_{kj} R_{k,t}, \qquad \phi_{kj} = \frac{n_k P_{kj}}{\sum_{k'} n_{k'} P_{k'j}}. \tag{3}$$

Products sent to OOS or abstained on are dropped. If nothing flows into an item, its weight is redistributed within its CPI group; true OOS products that leak into an item move with the official non-food index. Out-of-scope items are measured perfectly. The measured index is

$$\hat{I}_{A,t} = I_{A,t} + I_{A,L} \sum_{j \in A} \frac{w_j}{W_A} \big(\tilde{R}_{j,t} - R_{j,t}\big), \tag{4}$$

and the error is reported in basis points of month-on-month and year-on-year inflation for the headline, food and core indices. A perfect classifier gives $\tilde{R} = R$ and reproduces every official series exactly, which is a unit test of the code.

Decomposition. Since $\sum_k \phi_{kj} = 1$, $\tilde{R}_{j,t} - R_{j,t} = \sum_k \phi_{kj}(R_{k,t} - R_{j,t})$. The contribution of each confusion $k \to j$ is therefore the item weight times the misallocated mass times the gap between the two items' price movements. A confusion matters only if it is frequent, falls on a heavily weighted item and joins items whose prices move differently.

Uncertainty. The confusion counts come from the gold sample, with the Horvitz–Thompson universe shares n_k. We draw 1,000 confusion matrices from a Dirichlet posterior, row by row. Sparse rows need a prior, and a flat one is not neutral: it sends mass to arbitrary items and creates error even for a perfect classifier. We therefore use an empirical-Bayes prior worth $\alpha = 1$ product per row, whose mean puts the method's pooled accuracy on the diagonal and spreads the rest over the columns where that method's errors actually go. Under this prior a perfect classifier has zero error in every draw. Two variants bracket the result: a perfect classifier on the same universe (any error left is due to coverage alone) and the method's matrix plus product-level sampling noise, in which each product's log price drifts from its item's as a random walk with an assumed annual standard deviation of 5% or 10%, and item j averages the products sent to it.

4.4 The label ceiling (RQ4)

Treating the first labelling pass as the truth and the second as a classifier puts the human ceiling on the same scale as the models: we report agreement, Cohen's kappa (Cohen, 1960) and acc_w at the item, group and division level, with bootstrap intervals.

5 Results

[This part is written once the gold labels are evaluated.]

6 Discussion: what this means for central banks in Asia

[This part is written once the gold labels are evaluated.]

7 Limitations

One labeller. The gold labels are the author's. The second pass measures how consistent one person is with themselves, not how far two trained annotators would disagree, so the label ceiling of Section 5 is optimistic. Gold labels can also be wrong in ways that favour a model which shares the labeller's reading of the item headings.

The product universe is Open Food Facts. The gold frame is the Thai-script products in a crowd-sourced database, not the assortment of Thai shops. It over-represents packaged, branded products that volunteers chose to enter, and the universe shares n_k in the simulator are shares of that frame. A statistical office would draw the frame from its own scanner or web data, where the mix of items, and therefore the error in basis points, may differ.

Item weights below the published cells are estimated. TPSO publishes weights for its aggregates only. The equal split and the NNLS split are two transparent estimates; they agree on the cells but not on the items inside them. Results that depend on which item inside a cell carries the weight are reported under both.

The simulator measures classification error only. It takes official item relatives as the true price movements of every product and changes nothing but the assignment of products to items. A real web-price index would add sampling, matching, quality-adjustment and outlier error on top. Three modelling choices are assumptions rather than estimates: leaked out-of-scope products move with the official non-food index, an item that receives no products takes its group's measured movement, and the product-level dispersion of the noise variants is assumed, not measured.

Official specifications are not shop listings. The official labels come from TPSO's price-collection specifications. They are short and regular; shop and database texts are longer and noisier. The learning curve therefore measures the value of this kind of official label, not of labelled shop data.

Closed models change. The LLMs are reached through a commercial API. Model revisions, prices and refusal behaviour can change after the run date; every response is cached so that the numbers here can be reproduced, but not necessarily re-obtained. The official specifications are public and may have been seen in pre-training. Local compute is costed at an assumed price per GPU-hour.

Not comparable to Spectrum. Spectrum classifies euro-area web data into the ECOICOP 2018 sub-classes (BIS Innovation Hub, 2026). Our taxonomy, language, product universe, label source and metrics differ, so our numbers are not directly comparable with theirs, and we do not compare them.

8 Conclusion

[This part is written once the gold labels are evaluated.]

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A Data sources and reproducibility

The repository rebuilds every number in this paper from public sources. `make data` downloads the sources and checks their checksums; `make models` computes the embeddings, trains the classifiers and runs the LLMs (cached responses are replayed, so no API call is made when the cache is present); `make eval econ agreement` scores the methods on gold, runs the simulator and the label-ceiling analysis; `make figures paper` draws the figures and builds this PDF. `cpilens report` writes every number the paper prints into `paper/numbers.tex`, and a check in continuous integration fails if the paper's sources contain a number typed by hand. `results/run_manifest.json` records package versions, model revisions and the random seed (20261027).

B In-scope items and crosswalk

Table 2 lists the in-scope items with their COICOP 2018 classes and both estimated weights. The crosswalk from Open Food Facts category tags to these items is in `taxonomy/off_categories_to_thcpi.csv`, with the basis of each mapping (usually the TPSO item heading it follows).

C LLM prompt

The system prompt opens with the instructions below. It then lists every label by CPI group (code, Thai and English name, and what it includes) and the official examples, each as `text -> label`. The product is sent as the user message. The full prompt is generated by `cpilens.classify.llm.system_prompt`.

Table 1: Data sources. Retrieval dates and SHA-256 checksums of every file are in the repository (`data/SOURCES.md`, `data/manifest.csv`).

Source	Use	Licence
Open Food Facts, Open Beauty Facts (Open Food Facts contributors, 2026)	Product text; weak labels; gold frame	ODbL
TPSO open API (Trade Policy and Strategy Office, 2026)	Monthly item and aggregate indices	Thai open government data
TPSO monthly CPI release	Price-updated weights of the aggregates	Thai government publication
TPSO base-year note (Trade Policy and Strategy Office, 2025)	Division weights; base-year changes	Thai government publication
MOC open data (Ministry of Commerce, Thailand, 2026)	Official specifications (official labels)	Thai open government data
COICOP 2018 (United Nations Statistics Division, 2018)	Class codes of the in-scope items	Cited

Table 2: In-scope items. Weight: price-updated CPI weight, % of headline, estimated inside official weight cells by an equal split or by NNLS. Source: TPSO; UNSD; author's calculations.

Code	Item	COICOP 2018	Equal	NNLS
11110	Rice	01.1.1	1.16	2.57
11121	Flour	01.1.1	1.16	0.02
11122	Flour products (bread; noodles; pasta)	01.1.1	1.16	0.88
11211	Fresh meat	01.1.2	0.89	2.58
11212	Processed meat	01.1.2	0.89	0.39
11221	Fresh poultry	01.1.2	0.89	0.95
11231	Fresh freshwater fish	01.1.3	0.89	1.22
11232	Fresh marine fish	01.1.3	0.89	0.35
11233	Other fresh seafood	01.1.3	0.89	0.87
11234	Processed fish	01.1.3	0.89	0.62
11235	Processed seafood	01.1.3	0.89	0.13
11310	Eggs	01.1.4	0.88	0.81
11320	Milk and dairy products	01.1.4	0.88	0.95
11411	Fresh vegetables	01.1.7	2.13	2.13
11412	Processed vegetables and other	01.1.7	0.23	0.43
11421	Fresh fruit	01.1.6	2.27	2.27
11422	Processed fruit and other	01.1.6	0.23	0.03
11521	Oils and fats	01.1.5	0.59	0.44
11522	Seasonings and condiments	01.1.9	0.59	0.73
11600	Non-alcoholic beverages	01.2.1, 01.2.2, 01.2.3, 01.2.4, 01.2.5, 01.2.6, 01.2.9	3.29	3.29
11710	Sugar and confectionery	01.1.8	0.84	0.84
12300	Ready-to-eat food	01.1.9, 11.1.1	14.78	14.78
36000	Cleaning products	05.6.1	1.54	1.61
42100	Personal care products	13.1.2	0.43	3.37
71000	Tobacco products	02.3.0	0.31	0.43
72100	Beer	02.1.3	0.31	0.54
72200	Wine	02.1.2	0.31	0.01
72300	Spirits	02.1.1	0.31	0.26

```
You classify products sold in Thai shops into the item categories of the Thai consumer
 ↪ price index (CPI, base year 2023 = BE 2566), compiled by the Trade Policy and Strategy
 ↪ Office (TPSO). Each product is given as `name | brand | quantity`; names are usually
 ↪ Thai, sometimes English or other languages.

Pick exactly one label. First choose the group, then a label inside that group. Use OOS
 ↪ when the product belongs to none of the labels (for example pet food, medicines,
 ↪ kitchenware, stationery, electronics). Classify what the product is, not its flavour:
 ↪ a shrimp-flavoured cracker is a snack (11122), not seafood. Confidence is your
 ↪ probability, 0-100, that the label is correct.
```

D Hyperparameters

Table 3: Hyperparameters, as read from the code.

Setting	Value
k-NN: candidate k	1, 3, 5, 10, 20, 50
Linear and MLP: optimiser, learning rate, weight decay	Adam, 10^{-3}, 10^{-4}
MLP: hidden units, dropout	512, 0.2
Batch size (at most); early-stopping patience	512; 3
Fine-tuning: epochs (at most), batch, learning rate	4, 64, 3×10^{-5}
Fine-tuning: maximum input length (tokens)	64
LLM: official examples per item; maximum output tokens	2; 2,048
Cascade: candidate thresholds (the last routes every product)	0, 0.3, 0.5, 0.6, 0.7, 0.8, 0.85, 0.9, 0.95, 0.98, 1.01
Official-label budgets	100, 250, 500, 1,000 and all
Bootstrap resamples; simulator draws; prior weight α	1,000; 1,000; 1
Noise variants: annual dispersion (%)	5, 10
Gap between labelling passes (hours, at least)	48

E Compute and cost ledger

Every billed API call is logged with its tokens and cost in `results/llm_costs.csv`; Table 4 sums it. Input tokens include prompt-cache writes and reads, which are billed at different rates.

Table 4: LLM calls and cost (USD) by model and stage. Smoke and probe: pipeline checks; eval: classification of the evaluation and development sets. Source: author's API logs.

Model	Stage	Calls	Input tokens	Output tokens	USD
claude-haiku-5-5	eval	7,031	38,870,803	225,955	0.55
claude-haiku-5-5	smoke	5	27,665	121	0.00
claude-opus-5-5	eval	2,509	13,845,813	77,798	5.02
claude-opus-5-5	probe	10	55,304	262	0.05
Total		9,555	52,799,585	304,136	5.62